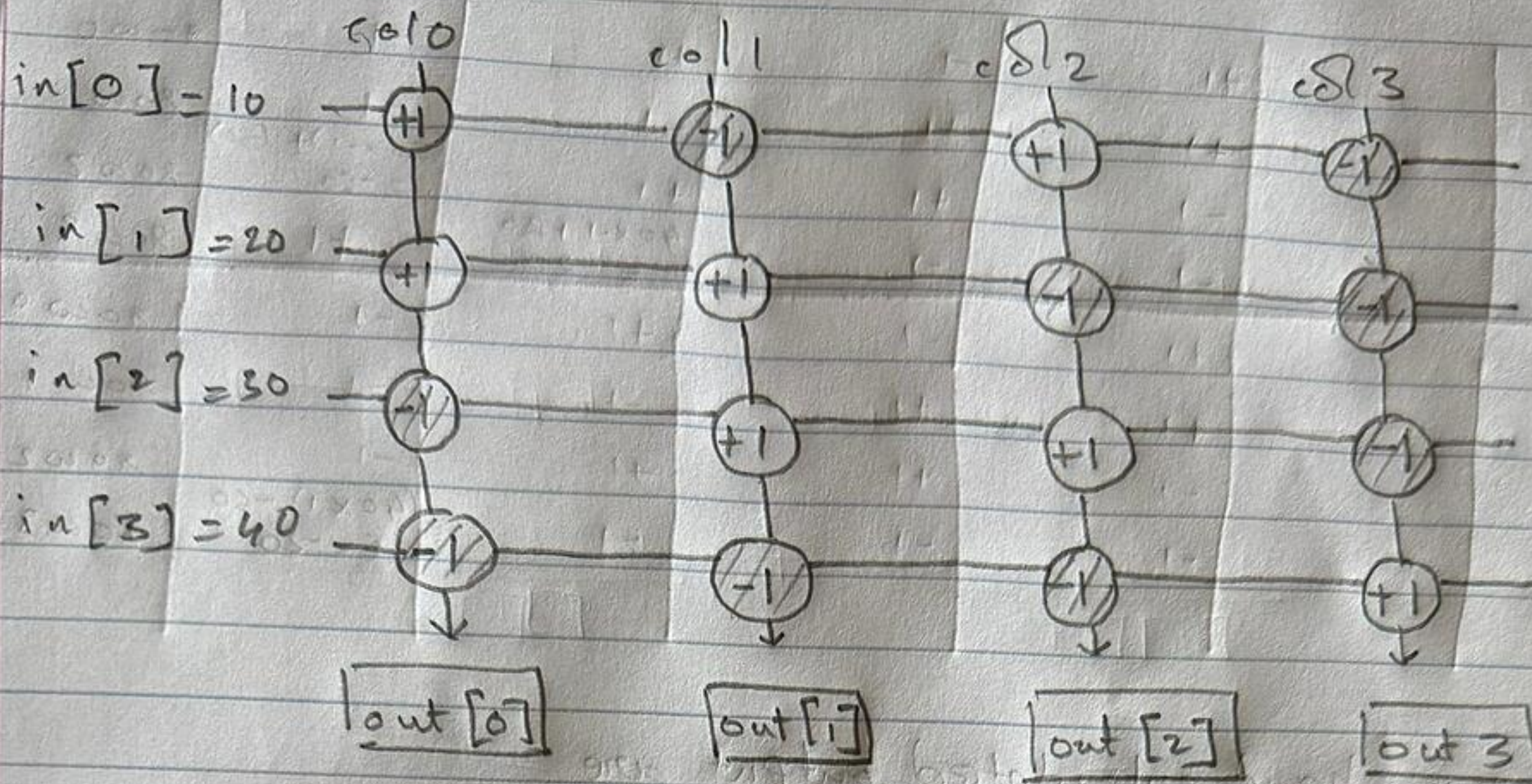




clk ticks	in	col 0	col 1	col 2	col 3	row	Output
t=0	10	$10 \times 1 = 10$	-1	+1	-1	row 0	
		+1	+1	-1	-1	row 1	
		-1	+1	+1	-1	row 2	
		-1	-1	-1	+1	row 3	
t=1	20	$10 \times -1 = -10$	+1	-1	-1	row 0	
		$(20 \times 1) + 10 = 30$	+1	-1	-1	row 1	
		-1	+1	+1	-1	row 2	
		-1	-1	-1	+1	row 3	
t=2	30	+1	-1	$10 \times 1 = 10$	-1	row 0	
		+1	$(20 \times 1) - 10 = 20$	-1	-1	row 1	
		$(30 \times -1) + 30 = 0$	+1	+1	-1	row 2	
		-1	-1	-1	+1	row 3	
t=3	40	+1	-1	+1	$10 \times -1 = -10$	row 0	out[0] = -40
		+1	+1	$(20 \times -1) + 10 = -20$	-1	row 1	
		-1	$(30 \times 1) + 10 = 40$	+1	-1	row 2	
		$(40 \times -1) + 0 = -40$	-1	-1	+1	row 3	
t=4		+1	-1	+1	-1	row 0	out[0] = -40
		+1	+1	-1	$(20 \times -1) - 10 = -30$	row 1	out[1] = 0
		-1	+1	$(30 \times 1) + (-10) = 20$	-1	row 2	
		-1	$(40 \times -1) + 40 = 0$	-1	+1	row 3	

t=5	+1	-1	+1	-1	row 0	out[0] = -40
	+1	+1	-1	-1	row 1	out[1] = 0
	-1	+1	+1	(30x-1) + (-30) = -60	row 2	out[2] = -20
	-1	-1	40x-1 + 20 = -20	+1	row 3	
t=6	+1	-1	+1	-1	row 0	out[0] = -40
	+1	+1	-1	-1	row 1	out[1] = 0
	-1	+1	+1	-1	row 2	out[2] = -20
	-1	-1	-1	(40x1) - 60 = -20	row 3	out[3] = -20

So, my hand calculated output are

$$\text{out}[0] = -40$$

$$\text{out}[1] = 0$$

$$\text{out}[2] = -20$$

$$\text{out}[3] = -20$$

And the simulation results are also the same.